

ADITYA PRAKASH

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SUMMARY

Software engineering student building AI-native and security-focused systems. Experienced with **C++, Python, TypeScript, Git/GitHub, and Linux**, with hands-on work in **multithreaded systems, networking, LLM-based applications, and vector search**. Comfortable working with **AI coding tools** (Claude and similar) — writing a clear brief, reviewing generated code, and owning what ships. Built systems spanning **deep packet inspection, security policy enforcement, and retrieval-augmented generation**. Interested in AI-native product engineering and reliable software built with human-reviewed AI assistance.

PROJECTS

Deep Packet Inspection (DPI) Engine | C++, Wireshark, Multithreading

- Developed a **C++17 Deep Packet Inspection engine** to parse Ethernet, IPv4, and TCP/UDP packets and analyze network traffic from PCAP captures.
- Implemented **stateful flow tracking** using five-tuples (source/destination IP, ports, protocol) to correlate packets belonging to the same network connection.
- Built **TLS SNI and HTTP Host inspection** to identify application/domain traffic (e.g. YouTube, Facebook, Google) without decrypting HTTPS payloads.
- Implemented **security policies** for IP, application, and domain-based traffic blocking, marking and filtering malicious/restricted flows.
- Designed a **multithreaded packet-processing pipeline** using load balancers, fast-path workers, and thread-safe producer-consumer queues for parallel traffic analysis.

RECALL DSA: RAG Pipeline for Educational Video | Python, MLX-Whisper, Qdrant, LangChain, Grok API

- Architected a **timestamp-aware Retrieval-Augmented Generation (RAG) system** to index and semantically search 50+ hours of DSA video lectures.
- Optimized local speech-to-text inference using **Apple's MLX framework** (Apple Metal GPU), bypassing CPU bottlenecks to transcribe complex technical audio.
- Engineered an **NLP data-cleaning pipeline** using LLM prompting to correct phonetic hallucinations in code-switched (Hinglish) domain jargon.
- Implemented **multilingual embeddings (BGE-M3)** stored in **Qdrant DB** with embedded URL metadata, letting users jump directly to the exact video timestamp of a concept.

SKILLS

Programming: C++, Python, TypeScript, SQL

Computer Science: DSA, OOP, DBMS

AI/ML & Data: LLM-based applications, RAG, Vector Search (Qdrant), LangChain, Machine Learning, Data Preprocessing, EDA

Networking: TCP/IP stack, DNS, HTTP/HTTPS, Stateful Flow Tracking

Tools & Platforms: Git, GitHub, Linux, Wireshark, Claude, VS Code

Languages: Hindi, English

EDUCATION

Manipal University Jaipur

Bachelor of Technology (B.Tech), Computer Science Engineering

Jaipur

July 2024 – July 2028

TRAINING/CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle (2025). Validated foundational knowledge of AI/ML concepts, generative AI, and OCI AI services.

Introduction to Data Engineering and Big Data — GUVI & HCL. Covered ETL pipelines, data warehousing, and big data processing tools.

Deloitte Australia — Data Analytics Job Simulation. Completed simulated data analytics tasks: data cleaning, forensic analysis, and stakeholder reporting.